

# Verification as a Costed Action: When Checking Beats Containing Under Attack–Fault Ambiguity

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## Abstract

Order flow alone cannot always tell whether a trading-gateway anomaly comes from a compromised account or a benign fault. A check such as the account holder’s confirmation through a separate channel reads authorization; if its outcome is conditionally independent of order flow given the cause, it is not bound by what order flow reveals, composes with any passive posterior and, when order flow is uninformative, lowers the cost of ambiguity exactly when the prior lies in a verification interval. The check has a sensitivity, a specificity, a cost and a delay. The interval is that of the classical two-threshold leave-verify-contain rule; because delay harm accrues only under compromise, it vanishes at a closed-form critical delay. The interval does not depend on the prior; the prior sets how much posterior mass reaches it. In the companion study’s cells where the fitted classifier does no better than the best constant policy at the point estimate, an exploratory evaluation of the rule on that study’s corrected posteriors, with the check’s characteristics set on a grid rather than measured, is mixed: its pre-stated agreement criterion is met in every such cell, but it counts mostly points where no saving is predicted, and restricted to the points where a saving is predicted it fails in 2 cells. Either reading checks the calibration of the corrected posteriors, not the proved rule or any real check. On provisional cells the test weakened the prediction.

*Keywords:* costly verification, value of information, threshold decision rule, intrusion response, attack–fault ambiguity, electronic trading, decision

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## 1. Introduction

An operator of an electronic trading gateway sees an account submit an unusual pattern of orders. The pattern may come from an adversary who has taken over the account or the session, or from a benign malfunction of the client’s own systems. The operator can *contain*—suspend the session, pull the orders—or *leave* trading running. Containing a fault disrupts a legitimate client for nothing; leaving a compromise lets the adversary keep trading. When the two causes produce order flow that looks the same, no amount of order-flow analysis settles the matter.

*The separate channel.* A companion study, Paper 1 [1], makes this precise. In its linear-Gaussian order-flow model, when the confounding condition holds, order-flow observation cannot separate a compromise from a fault; the statements of Paper 1 used here (its Theorem 1, Definitions D5 and D6, and Propositions 1 and 2) are restated in the Supplementary Material, Section S2. By Paper 1’s Proposition 1, an operator who sees only order flow can then do no better than act on the prior, and pays a cost of ambiguity, over an operator who knows the cause, equal to the expected regret of the better of the two terminal actions taken at the prior (for our loss table, Corollary 1(i)). Verification reads another channel, *authorization*, and, if its outcome is conditionally independent of order flow given the cause (Assumption 2), it is not subject to that limit; Section 3.4 develops the argument.

*When a check pays.* A check is not free. It costs analyst time [2] and client friction [3], and while it runs a real adversary keeps trading. We model verification as an action with sensitivity  $s$ , specificity  $t$ , a fixed cost  $c_v$  and a delay  $\tau$  during which a true compromise causes damage at rate  $r_K$ , on top of a general loss table  $(C_K, C_F, H_K, H_F)$ . Comparing the value of a check, a tent peaking at the passive threshold  $q^*$  (Lemmas 1 and 3), with its cost gives the classical leave / verify / contain rule of Pauker and Kassirer [4] (Theorem 1), restated, not claimed, because every other result is read against its thresholds  $q_1 < q^* < q_2$ . What the setting adds is the shape of the cost: harm during the delay accrues only under compromise, so the expected cost of checking,  $c_v + q r_K \tau$ , grows with the belief, and the interval  $(q_1, q_2)$  vanishes at a critical delay  $\tau^*$  (Theorem 2), a time budget beyond which a check should never be run, at any belief.

*Contributions.*

1. *The separate channel.* If the outcome of a check that reads authorization is conditionally independent of order flow given the cause (Assumption 2, a modelling assumption, not a result), the check composes with any passive posterior (Lemma 2) and lowers the confounded cost of ambiguity exactly when the prior lies in the verification interval (Corollary 1). The confounded case is then the worst case for the passive and the verify-first operator alike (Corollary 2).
2. *The cause-borne delay.* A closed-form critical delay  $\tau^*$  at which the verification interval vanishes, with nested intervals below it and the manner of vanishing characterised (Theorem 2), under a cost of delay borne only by a true compromise. The direction of the bound depends on how the delay is charged:  $\tau^*$  is a lower bound on the critical delay when the cost of leaving a compromise already includes part of the delay harm (Corollary 3), and an upper bound when a fault also costs during the check (Corollary 4).
3. *What the prior does.* The interval does not depend on the prior, but the posterior mass that reaches it vanishes as compromise becomes rare, except for a free check that never flags a fault, where its limit is positive exactly when the order-flow laws of compromise and fault are not mutually singular (Lemma 4(iii)–(iv)).
4. *An exploratory evaluation of the rule on the companion study’s corrected posteriors,* on its market designs and priors (Section 5), with the grid of  $(s, t, c_v, \tau)$ , the plausible set, the five policies and the primary test’s outcome criterion fixed before any quantity of the primary test was computed. Both outcomes are exploratory. In the cells where passive classification does not beat the best constant policy at the point estimate, targeted verification beats that policy at plausible grid points; the primary test reads *supports* under its pre-stated criterion, which counts mostly points where no saving is predicted, and *weakens* in 2 cells under a post-hoc reading restricted to the points where a saving is predicted; either reading checks the calibration of the corrected posteriors. After family-wise control, the policy comparison is *inconclusive at this power*. No pre-registered confirmation has been run.

Three supporting extensions are not headline results, and the paper does not rest on them: Proposition 2 (Section 6.4) states when targeted verification on a miscalibrated score still beats the passive rule, a candidate mechanism

not tested here (its remedy half is known from decision calibration); Proposition 3 (Section 6.2) replaces Assumption 2 by lower bounds on the check’s accuracy; and Theorem 3, stated only in the Supplementary Material, carries the interval and the critical delay over to a check with any output through its total variation (it follows in one line from known results).

Sections 2–7 give the related work, the model, the results, the exploratory study, the discussion and the conclusion. The Supplementary Material contains the proofs, the restated results of Paper 1, the full empirical design, the secondary tables, and the statements of Remarks 1 and 2, Proposition 1, Definition 6 and Theorem 3.

## 2. Related work

*Threshold analysis and the value of information.* The leave / verify / contain rule of Section 4 is the threshold approach of Pauker and Kassirer [4] transposed to attack–fault triage: a *testing threshold* and a *test-treatment threshold*, set by the test’s sensitivity, specificity and cost and the benefit and harm of treatment, split the probability of disease into withhold treatment, test, and treat without testing. Our  $q_1$  and  $q_2$  play the same roles, with contain in place of treat. The underlying calculation—a single experiment is worth running exactly when its expected value of sample information exceeds its cost, and for two actions and two states that value is piecewise linear—is classical [5]. The largest proportional reduction of expected regret a test achieves is its Youden index  $s + t - 1$ , reached at the treatment threshold [6]. Three-way decisions derive accept / defer / reject thresholds from a loss table, and the deferral region vanishes when the two thresholds cross [7]; deferral there carries no information, so it gives a critical delay for waiting without a test, not for a check with a sensitivity and a specificity.

*Costs of delay that depend on the state.* The feature that distinguishes our setting is a cost of checking that falls on one cause: an adversary keeps trading while the check runs, while a fault is taken not to get worse (Assumption 4, relaxed in Corollary 4). Time-critical decision analysis computes the expected cost of deferring an action from a distribution over states and a time-dependent utility [8]. In medical decision making, Schroeder et al. [9] use the threshold framework to trade the accuracy of a slower send-out test against the clinical benefit lost while waiting for its result; this is the closest precedent we know for the delay term  $qr_K\tau$  of our model. In operations,

belief-threshold diagnosis under congestion charges the delay of acquiring information to the queue rather than to one cause [10]. Neither the rule, nor the tent shape of the value of information (Lemma 3), nor the critical delay is new as decision theory; what is new here is the channel and the cause-borne delay: their use for attack-fault triage, where the cost of delay falls only on a compromise, and the separate-channel argument of Section 3.4.

*Costly and active information acquisition.* Costly observation actions in partially observed decision processes go back to Monahan [11]; ours is the one-shot case with two causes, two terminal actions and one check, which gives the thresholds and the critical delay in closed form. In its sequential analogue, the Bayesian sequential probability ratio test, the continuation interval shrinks as the cost of sampling rises [12], but that cost does not depend on the cause.

*Intrusion response and alert verification.* Intrusion response has been cast as a cost-sensitive decision [13]. Alert verification checks whether an intrusion attempt succeeded before an alert is raised [14], with no cost, delay or threshold attached. Investigating an alert has been modelled as costly information acquisition [15]; deferring uncertain alerts to an analyst is learned in security-operations triage [16]; and a two-threshold rule escalates high scores, closes low ones and defers the rest [17]. There the thresholds come from error-rate budgets or are learned; the cost of the check does not grow with the belief, and there is no critical delay. Separating cyberattacks from physical faults has been studied by adding cyberspace logs to physical measurements, an analysis that in industry is manual and slow, which that work sets out to shorten without pricing its time [18]. This is the closest precedent we know for reading a second channel; what it does not give is a rule for when reading that channel is worth its cost and delay.

### 3. Model

An operator watches an anomaly on a trading gateway caused either by a compromise  $K$  or by a fault  $F$  (Definition 1). The operator can *contain* (suspend the account or session, pull the orders) or *leave* (let trading continue) and, before either, may *verify*: query a source of evidence that the order flow does not carry.

### 3.1. Loss table and passive risk

The operator's belief that the cause is a compromise is  $q \in [0, 1]$ . Without verification, the operator picks the terminal action with the lower expected loss.

**Definition 1 (Loss table, passive risk).** There are two causes, compromise  $K$  (the orders are submitted through a valid credential or session by someone the holder of that credential or session did not authorize: a stolen credential, a hijacked session) and fault  $F$  (the orders are the holder's own, distorted by a benign malfunction), and two terminal actions, *contain* and *leave*. Containing costs  $C_K$  under  $K$  and  $C_F$  under  $F$ ; leaving costs  $H_K$  under  $K$  and  $H_F$  under  $F$ . For a belief  $q = P(K) \in [0, 1]$  the *passive risk* is

$$R(q) = \min\{q C_K + (1 - q) C_F, q H_K + (1 - q) H_F\}.$$

Write  $a = H_K - C_K$  and  $b = C_F - H_F$ .

Containing a compromise ( $C_K$ ) costs the disruption of a legitimate client plus the operational cost of the response; containing a fault ( $C_F$ ) costs the same disruption with nothing gained; leaving a compromise ( $H_K$ ) costs the harm the adversary goes on to do; leaving a fault ( $H_F$ ) costs what the malfunction costs if it is not interrupted, counted from the terminal decision on. The quantity  $a$  is what containment saves under a compromise, and  $b$  what it wastes under a fault. The table of Paper 1's toy scenario [1] is the special case  $H_F = 0$  of its general table (its Definition D6), with Paper 1's containment cost, disruption and harm in the roles of  $C_K$ ,  $C_F$  and  $H_K$ ; the cost table on which Paper 1 classifies its cells uses, in addition, a measured cost of leaving a fault, which differs by design and regime (Section 5.1).

### 3.2. Verification

On a trading gateway, verification is for example a call from the desk to the client to confirm the activity, or a step-up authentication that asks the submitter for a factor bound to the account holder.

**Definition 2 (Verification action).** Verification returns an outcome  $Z \in \{+, -\}$  with sensitivity  $s = P(Z = + | K)$  and specificity  $t = P(Z = - | F)$ . It costs  $c_v$  and takes a delay  $\tau$ ; during the delay a true compromise causes harm at rate  $r_K$ . From belief  $q$ ,

$$P(+) = qs + (1 - q)(1 - t), \quad P(-) = q(1 - s) + (1 - q)t,$$

$$q^+ = \frac{qs}{P(+)}, \quad q^- = \frac{q(1 - s)}{P(-)}$$

(a posterior whose outcome has probability zero is never used; see Definition 3).

The cost  $c_v$  is paid whatever the cause (analyst time, the call, client friction); the damage rate  $r_K$  is the harm per unit of delay done by an adversary who is really present, distinct from the cost  $H_K$  of leaving a compromise.

**Definition 3 (Verify-first risk, value of information, verification region).** The *verify-first risk* is

$$R_v(q) = c_v + q r_K \tau + P(+ ) R(q^+) + P(- ) R(q^-),$$

where  $P(z) R(q^z)$  is read as  $\min\{P(z, K) C_K + P(z, F) C_F, P(z, K) H_K + P(z, F) H_F\}$  with  $P(z, K) = P(z \mid K) q$  and  $P(z, F) = P(z \mid F) (1 - q)$  (equal to the product when  $P(z) > 0$ , and zero when  $P(z) = 0$ ). The *value of information* is  $V(q) = R(q) - [P(+ ) R(q^+) + P(- ) R(q^-)]$ , the *verification cost* is  $\chi(q) = c_v + q r_K \tau$ , and the *verification region* is

$$S = \{q \in [0, 1] : R_v(q) < R(q)\} = \{q \in [0, 1] : V(q) > \chi(q)\}.$$

Ties are resolved in favour of the passive action: the operator verifies only when verifying is strictly cheaper.

Because harm during the delay accrues only under  $K$ , the verification cost  $\chi$  grows with the belief: checking is more expensive precisely when a compromise is more likely.

### 3.3. Assumptions

**Assumption 1 (Cost orderings).**  $H_K > C_K$  and  $C_F > H_F$ ; that is,  $a > 0$  and  $b > 0$ : leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it.

Assumption 1 excludes the cases in which one terminal action dominates whatever the cause;  $a, b < 0$  is the same problem with the actions exchanged (Remark 1(b), Supplementary Material, Section S1). On a gateway it says that an unchecked adversary does more damage than a response costs, and that suspending a healthy client costs more than letting a benign fault run.

**Assumption 2 (Separate channel).** Verification reads authorization, not order flow: it asks whether the holder of the credential or session authorized the orders, through a channel the person submitting them does not control (for example an out-of-band confirmation with the account holder, or a step-up authentication with a factor bound to the holder). A check of whether the credential or session is valid is not such a check, since a

stolen credential or a hijacked session is valid under  $K$ . Its outcome  $Z$  is conditionally independent of the order-flow observation  $Y$  given the cause:  $P(Z, Y | c) = P(Z | c) P(Y | c)$  for  $c \in \{K, F\}$ , and  $(s, t)$  do not depend on  $Y$ .

**Assumption 3 (Informative one-shot verification).**  $s, t \in [0, 1]$  with  $s + t > 1$ ;  $c_v \geq 0$ ,  $\tau \geq 0$ ,  $r_K \geq 0$ . Verification is available once; after its outcome the operator takes a terminal action (contain or leave) and cannot verify again.

A check with  $s + t = 1$  carries no information (Remark 1); one with  $s + t < 1$  is a check with  $s + t > 1$  whose two outcomes have been relabelled. The one-shot restriction matches a triage step in which one check is run and the operator then acts; sequential checking is a stopping problem outside this paper.

**Assumption 4 (Delay accounting).** The harm  $r_K \tau$  accrued by a true compromise during the verification delay is *additional* to the terminal cost ( $C_K$  or  $H_K$ ) incurred afterwards, whichever terminal action is taken; the terminal costs  $C_K, C_F, H_K, H_F$  and the verification characteristics  $s, t$  do not depend on  $\tau$ ; a fault accrues no additional cost during the delay (its damage rate is  $r_F = 0$ ); and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 4 charges the delay harm in full on top of the terminal cost. Any double counting is therefore against verification only: when  $H_K$  already includes part of that harm, the critical delay  $\tau^*$  of Theorem 2 is a lower bound (Corollary 3). A fault that costs during the check is Corollary 4.

### 3.4. When verification is not bound by the order-flow limit

In its linear-Gaussian order-flow model, when the confounding condition holds, Paper 1 [1] shows that order-flow observation cannot separate a compromise from a fault, and that a passive operator who sees only order flow then pays a cost of ambiguity that no classifier can remove. That limit is a statement about the law of the order-flow observation  $Y$  under the two causes; it constrains every decision rule that is a function of  $Y$  alone. It does not constrain evidence that is not a function of  $Y$ .

Verification is such evidence: it reads the authorization channel, whether the entity submitting the orders is the one entitled to submit them. Assumption 2 states that, given the cause, the outcome of that check is independent of what the order flow shows, and that its accuracy  $(s, t)$  does not depend on the order flow: whether the account holder confirms the orders, or the

submitter passes a challenge bound to the holder, does not depend on how the orders look. The check does not detect a holder who trades maliciously on their own account, nor an adversary who also controls the holder's second factor; the model leaves both out of  $K$ . Under this assumption the joint posterior is the verification update applied to the order-flow posterior (Lemma 2), including when the order-flow laws under  $K$  and  $F$  coincide; that  $(s, t)$  do not depend on the order-flow law is Assumption 2 itself, not a finding of the lemma. That is the case in which Paper 1's passive operator can do no better than act on the prior; Corollary 1 states when verification brings the cost of ambiguity strictly below Paper 1's value there. Assumption 2 is a modelling choice, not a consequence of the setting; Section 6.2 states what is lost when it fails.

#### 4. The two-threshold rule and the critical delay

Lemmas 1 and 3 and Theorem 1 are background, stated because every other result is read against them; Theorem 2 gives the critical delay. Corollaries 1–4 apply these results to the confounded case, to every order-flow model and to the accounting of the delay, and Lemma 4 turns to a population of cases, which is where the prior enters. Propositions 2 and 3 are stated in Section 6, where they are used. All proofs, and the statements of Remarks 1–2, Proposition 1, Definition 6 and Theorem 3, are in the Supplementary Material, Section S1.

##### 4.1. The passive threshold

**Lemma 1 (Passive threshold).** Under Assumption 1, let  $q^* = b/(a + b) \in (0, 1)$ . Leaving is strictly optimal for  $q < q^*$ , containing is strictly optimal for  $q > q^*$ , and

$$R(q) = q H_K + (1 - q) H_F - (a + b) (q - q^*)^+,$$

so  $R$  is continuous, concave and piecewise linear on  $[0, 1]$  with a single kink at  $q^*$ .

The threshold  $q^*$  is the share of the total stakes  $a + b$  that a wrong containment wastes: a response that is costly to a healthy client pushes it up; an adversary who does much damage if left alone pulls it down.

#### 4.2. Verification composes with the passive posterior

**Lemma 2 (Verification composes with any passive posterior).**

Under Assumption 2, let  $q(y) = P(K \mid Y = y)$  be the passive (order-flow) posterior. Then  $P(K \mid Y = y, Z = z)$  equals the verification update of Definition 2 applied to  $q = q(y)$ : it is  $q(y)^+$  if  $z = +$  and  $q(y)^-$  if  $z = -$ . By Assumption 2, the characteristics  $(s, t)$  do not depend on the order-flow law; hence  $V$ ,  $\chi$  and  $S$  of Definition 3, which are built from  $q$ ,  $(s, t)$  and the constants alone, are the same whatever the order-flow law, including when the order-flow laws under  $K$  and  $F$  coincide, and the order-flow law enters only through the belief  $q(y)$  at which they are evaluated.

The operator first reduces the order-flow evidence to a belief  $q(y)$ , with whatever classifier or model is available, and then applies the verification analysis to that belief. All results below are stated for a belief  $q$  and hold at  $q = q(y)$ .

#### 4.3. The value of information is a tent

**Lemma 3 (The value of information is a tent).** Under Assumptions 1 and 3,

$$V(q) = \max\{0, \min\{G^+(q), G^-(q)\}\},$$

where

$$G^+(q) = q s a - (1 - q)(1 - t) b, \quad G^-(q) = (1 - q) t b - q(1 - s) a.$$

Consequently, with

$$q_L = \frac{(1 - t) b}{s a + (1 - t) b}, \quad q_U = \frac{t b}{t b + (1 - s) a},$$

(i)  $0 \leq q_L < q^* < q_U \leq 1$ ; (ii)  $V(q) > 0$  if and only if  $q \in (q_L, q_U)$ , equivalently if and only if  $q^- < q^* < q^+$  (the two posteriors fall strictly on opposite sides of  $q^*$ ); (iii)  $V$  is affine and strictly increasing on  $[q_L, q^*]$ , affine and strictly decreasing on  $[q^*, q_U]$ , and zero outside  $(q_L, q_U)$ ; (iv)  $V$  attains its maximum uniquely at  $q^*$ , with

$$V(q^*) = (s + t - 1) \frac{a b}{a + b} = (s + t - 1) a q^* > 0.$$

Lemma 3 is the standard calculation [5, 6]. Outside  $(q_L, q_U)$  no outcome can overturn the passive decision; the tent is not symmetric about  $q^*$  in general (Figure 1), and its height is the Youden index  $s + t - 1$  times the expected regret  $a q^* = b(1 - q^*)$  of either terminal action at the threshold.

#### 4.4. The two-threshold rule

**Theorem 1 (Two-threshold leave / verify / contain rule).** Under Assumptions 1, 3 and 4, the verification region  $S$  is an open interval (possibly empty). It is non-empty if and only if

$$V(q^*) > c_v + q^* r_K \tau,$$

and then  $S = (q_1, q_2)$  with  $q_L \leq q_1 < q^* < q_2 \leq q_U$ , where

$$q_1 = \frac{c_v + (1-t)b}{s a + (1-t)b - r_K \tau}, \quad q_2 = \frac{t b - c_v}{t b + (1-s)a + r_K \tau}$$

(both denominators are positive when  $S \neq \emptyset$ ). If  $S \neq \emptyset$ , the optimal one-shot policy is: *leave* if  $q \leq q_1$ , *verify* if  $q_1 < q < q_2$  and then contain on  $Z = +$  and leave on  $Z = -$ , *contain* if  $q \geq q_2$ . If  $S = \emptyset$ , write  $q_1 = q_2 = q^*$ ; the optimal policy is the passive rule of Lemma 1, *leave* if  $q < q^*$  and *contain* if  $q > q^*$ , and at  $q = q^*$ , where both terminal actions are optimal, the operator leaves by convention. When  $c_v = 0$  and  $\tau r_K = 0$ ,  $S = (q_L, q_U)$ .

Theorem 1 is the threshold rule of Pauker and Kassirer [4] (Section 2); the setting adds only a cost of the check that grows with the belief, through  $q r_K \tau$ . The rule is drawn in Figure 1. The interval contains  $q^*$  when it is non-empty, and its ends move inward as the check becomes more expensive or slower.

#### 4.5. The critical delay

**Theorem 2 (Critical delay).** Under Assumptions 1, 3 and 4, write  $S(\tau)$  for the verification region at delay  $\tau$ , all other parameters fixed. (i) *Nesting.*  $\tau < \tau'$  implies  $S(\tau') \subseteq S(\tau)$ . If  $r_K > 0$  and  $V(q^*) > c_v$ , then  $S(\tau) \neq \emptyset$  exactly for  $\tau \in [0, \tau^*)$  (part (ii)), and on  $[0, \tau^*)$  the end point  $q_2(\tau)$  of Theorem 1 is strictly decreasing in  $\tau$ , and  $q_1(\tau)$  is strictly increasing in  $\tau$  unless  $c_v = 0$  and  $t = 1$ , in which case  $q_1(\tau) = 0$  on  $[0, \tau^*)$ . For  $\tau \geq \tau^*$ ,  $S(\tau) = \emptyset$  and the convention  $q_1 = q_2 = q^*$  of Theorem 1 applies. (ii) *Critical delay.* If  $r_K > 0$  and  $V(q^*) > c_v$ , let

$$\tau^* = \frac{V(q^*) - c_v}{q^* r_K} = \frac{(s+t-1)a - c_v/q^*}{r_K} > 0.$$

Then  $S(\tau) \neq \emptyset$  if and only if  $\tau < \tau^*$ . (iii) *Degenerate cases.* If  $V(q^*) \leq c_v$ , then  $S(\tau) = \emptyset$  for every  $\tau \geq 0$ . If  $r_K = 0$ ,  $S(\tau)$  does not depend on  $\tau$ . (iv)

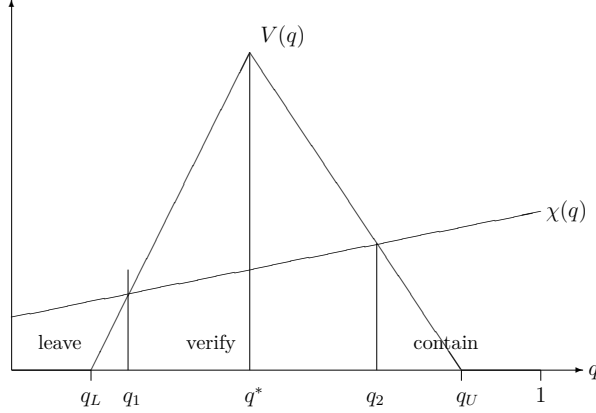


Figure 1: Schematic of Lemma 3 and Theorem 1 (not to scale; critical delays of the empirical study’s checks: Supplementary Material, Table S7). The value of information  $V$  is a tent on  $(q_L, q_U)$  with its peak at the passive threshold  $q^*$ ; the verification cost  $\chi$  increases with the belief because harm during the delay accrues only under compromise. Verification pays on  $(q_1, q_2)$ , where the tent lies above the cost line. Raising the delay  $\tau$  tilts the cost line upward and shrinks the interval (Theorem 2).

*How the interval vanishes.* Suppose  $r_K > 0$  and  $V(q^*) > c_v$  (the hypotheses of (ii)). If  $c_v > 0$  or  $t < 1$ , then  $q_1 \rightarrow q^*$  and  $q_2 \rightarrow q^*$  as  $\tau \uparrow \tau^*$ : the interval shrinks continuously to the point  $q^*$ . If  $c_v = 0$  and  $t = 1$ , then  $S(\tau) = (0, q_2(\tau))$  for  $\tau < \tau^*$  with  $q_2(\tau) \downarrow q^*$ , and  $S(\tau^*) = \emptyset$ : the interval vanishes discontinuously.

Because the tent peaks at  $q^*$  and the cost of the check is affine and increasing in the belief, the check pays at some belief if and only if it pays at  $q^*$ ; solving that inequality for the delay gives  $\tau^*$ . For an operator,  $\tau^*$  is a time budget: a procedure that takes longer should never be run, at any belief. The budget grows with the accuracy of the check and shrinks with the damage rate and the fixed cost; if  $c_v \geq V(q^*)$ , no delay is short enough (Theorem 2(iii)).

In the units of the empirical study (Section 5.2), the critical delay is  $r_K \tau^* / a = s + t - 1 - x$ , with  $x$  the fixed cost of the check as a share of the peak value of a perfect, free and instantaneous check. A worked example on the cost table of Paper 1 is in the Supplementary Material, Section S5.

#### 4.6. Comparison with the passive cost of ambiguity

To compare verification with the passive analysis of Paper 1 [1], the decision problem is stated over policies that act on the order-flow observation.

**Definition 4 (Decision problem with verification; verify-first cost of ambiguity).** The cause  $M$  is  $K$  with prior probability  $p = P(K) \in (0, 1)$  and  $F$  otherwise. Given  $M$ , the order-flow observation  $Y$  has law  $P_M^Y$ , and the verification outcome  $Z$  has the law of Definition 2, jointly as in Assumption 2. Write  $L(\text{contain}, K) = C_K$ ,  $L(\text{contain}, F) = C_F$ ,  $L(\text{leave}, K) = H_K$ ,  $L(\text{leave}, F) = H_F$  (Definition 1). A *verify-first policy* is a quadruple of measurable maps  $\delta, \pi_0, \pi_+, \pi_-$  from observations  $y$  to  $[0, 1]$ : with probability  $\delta(y)$  the operator verifies and then contains with probability  $\pi_Z(y)$ ; otherwise it contains with probability  $\pi_0(y)$ . Its expected cost is

$$\begin{aligned} J(\delta, \pi) = \mathbb{E} \bigg[ & (1 - \delta(Y)) (\pi_0(Y) L(\text{contain}, M) \\ & + (1 - \pi_0(Y)) L(\text{leave}, M)) \\ & + \delta(Y) (c_v + \mathbb{I}[M = K] r_K \tau + \pi_Z(Y) L(\text{contain}, M) \\ & + (1 - \pi_Z(Y)) L(\text{leave}, M)) \bigg]. \end{aligned}$$

A *passive policy* is a verify-first policy with  $\delta \equiv 0$ . Let  $R_Y^*$  and  $R_{Y,v}^*$  be the infima of  $J$  over passive and over verify-first policies,  $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$  the expected cost of an operator who observes  $M$ , and

$$\text{CoA}^* = R_Y^* - R_{\text{oracle}}, \quad \text{CoA}_v^* = R_{Y,v}^* - R_{\text{oracle}}$$

the passive and the verify-first cost of ambiguity.

**Corollary 1 (Verification lowers the confounded cost of ambiguity).** Under Assumptions 1–4, in the setting of Definition 4, suppose the order-flow laws coincide,  $P_K^Y = P_F^Y$  (the confounded case). Then: (i)  $\text{CoA}^* = \min\{p a, (1 - p) b\} > 0$ ; (ii) the verify-first cost of ambiguity is

$$\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\},$$

$$E(p) = c_v + p r_K \tau + p(1 - s) a + (1 - p)(1 - t) b,$$

equivalently  $\text{CoA}_v^* = \text{CoA}^* - (V(p) - \chi(p))^+$ , and it is attained by the policy of Theorem 1 applied at the belief  $q = p$ , whatever  $Y$ ; (iii)  $\text{CoA}_v^* < \text{CoA}^*$  if and only if  $p \in S$ , that is, if and only if  $V(q^*) > c_v + q^* r_K \tau$  and  $q_1 < p < q_2$  (Theorem 1); if  $r_K > 0$  and  $V(q^*) > c_v$ , this holds if and only if  $\tau < \tau^*$  and  $q_1(\tau) < p < q_2(\tau)$  (Theorem 2), and if  $V(q^*) \leq c_v$  it holds for no  $p$  and no  $\tau$ ; (iv) over  $p \in (0, 1)$ , the reduction  $\text{CoA}^* - \text{CoA}_v^*$  is largest at  $p = q^*$ , where it

equals  $(V(q^*) - c_v - q^* r_K \tau)^+$ ; (v)  $\text{CoA}_v^* = 0$  if and only if  $s = t = 1$ ,  $c_v = 0$  and  $r_K \tau = 0$ . Under the special case  $H_F = 0$  of Paper 1's Definition D6,  $\text{CoA}^*$  in (i) is the cost of ambiguity of Paper 1, Proposition 1, so (iii) states when verification brings the cost of ambiguity strictly below that value.

In the confounded case the order flow carries no information about the cause, and the passive cost of ambiguity in part (i), stated for the general loss table, reduces to the value of Paper 1's Proposition 1 when  $H_F = 0$ . Corollary 1 states when verification brings it strictly lower: exactly when the prior lies in the verification region of Theorem 1, so, for a harmful compromise and a check whose fixed cost is below the peak value of information, only when the check is faster than  $\tau^*$ .

The confounded case is the worst case for the passive and the verify-first operator alike:

**Corollary 2 (Uniform ceiling: the confounded case is the worst case).** Under Assumptions 1–4, in the setting of Definition 4, let  $E(p)$  be as in Corollary 1. For every pair of order-flow laws  $P_K^Y, P_F^Y$  and every  $p \in (0, 1)$ : (i) the *verify-everything* policy ( $\delta \equiv 1$ ,  $\pi_+ \equiv 1$ ,  $\pi_- \equiv 0$ ) has expected cost  $R_{\text{oracle}} + E(p)$ , whatever the order-flow laws; (ii)  $\text{CoA}^* \leq \min\{p a, (1 - p) b\}$  and

$$\text{CoA}_v^* \leq \min\{\text{CoA}^*, E(p)\} \leq \min\{p a, (1 - p) b, E(p)\};$$

(iii) every inequality in (ii) is an equality in the confounded case  $P_K^Y = P_F^Y$  (Corollary 1), so that, over all pairs of order-flow laws, the largest passive cost of ambiguity is  $\min\{p a, (1 - p) b\}$  and the largest verify-first cost of ambiguity is  $\min\{p a, (1 - p) b, E(p)\}$ , both attained in the confounded case; (iv) the largest verify-first cost of ambiguity is strictly below the largest passive one if and only if  $p \in S$ , and the gap is  $(V(p) - \chi(p))^+$ . Under the special case  $H_F = 0$  of Paper 1's Definition D6 and the hypotheses of Paper 1, Proposition 1,  $\min\{p a, (1 - p) b\}$  is the cost of ambiguity of Paper 1, Proposition 1, and the passive bound in (ii) is the upper bound of Paper 1, Proposition 2 (total-variation extension of Proposition 1): the value of Paper 1's Proposition 1 is the worst case of the passive cost of ambiguity over all order-flow models, and verification caps the cost of ambiguity at  $E(p)$  for every order-flow model.

**Corollary 3 (Delay accounting:  $\tau^*$  is a lower bound).** For  $\theta \in [0, 1]$ , *Assumption 4 $_\theta$*  is Assumption 4 except that the harm  $r_K \tau$  accrued by a true compromise during the delay is added to the terminal cost in full if the

terminal action is *contain* and in the share  $\theta$  if it is *leave*. Assumption 4<sub>1</sub> is Assumption 4; Assumption 4<sub>0</sub> is the accounting in which  $H_K$  already includes the harm a compromise left in place causes during the delay. Under Assumption 4<sub>0</sub> the verify-first risk is

$$R_v^\theta(q) = c_v + \sum_{z \in \{+, -\}} \min \left\{ P(z, K) (C_K + r_K \tau) + P(z, F) C_F, \right. \\ \left. P(z, K) (H_K + \theta r_K \tau) + P(z, F) H_F \right\},$$

with  $P(z, K)$ ,  $P(z, F)$  as in Definition 3, and the verification region is  $S_\theta(\tau) = \{q \in [0, 1] : R_v^\theta(q) < R(q)\}$ . Let  $\sigma_\theta = s + \theta(1 - s) \in [s, 1]$ . Under Assumptions 1, 3 and 4<sub>0</sub>: (i)  $S_\theta(\tau) = \{q \in [0, 1] : V(q) > c_v + \sigma_\theta q r_K \tau\}$ , the verification region of Definition 3 with  $r_K$  replaced by  $\sigma_\theta r_K$ ; for  $q \in S_\theta(\tau)$  the optimal action after verification is to contain on  $Z = +$  and leave on  $Z = -$ . Theorems 1 and 2 therefore hold under Assumption 4<sub>0</sub> with  $r_K$  replaced by  $\sigma_\theta r_K$ . (ii) For  $0 \leq \theta \leq \theta' \leq 1$  and every  $\tau \geq 0$ ,  $S(\tau) = S_1(\tau) \subseteq S_{\theta'}(\tau) \subseteq S_\theta(\tau) \subseteq S_0(\tau)$ . (iii) If  $r_K > 0$  and  $V(q^*) > c_v$ , then  $S_\theta(\tau) \neq \emptyset$  if and only if  $\tau < \tau_\theta^*$ , where

$$\tau_\theta^* = \frac{\tau^*}{\sigma_\theta}, \quad \tau^* \leq \tau_\theta^* \leq \frac{\tau^*}{s},$$

with  $\tau^*$  of Theorem 2. So  $\tau^*$  is a lower bound on the critical delay under every accounting 4<sub>0</sub>, attained if and only if  $\theta = 1$  or  $s = 1$ .

**Corollary 4 (A fault that accrues cost during the delay).** For  $r_F \geq 0$ , Assumption 4<sub>F</sub> is Assumption 4 except that a fault also accrues cost at rate  $r_F$  during the verification delay, additional to the terminal cost whichever terminal action is taken (Assumption 4 is the case  $r_F = 0$ ). Under Assumption 4<sub>F</sub> the verify-first risk is  $R_v(q) + (1 - q) r_F \tau$ , so the verification region is that of Definition 3 with  $\chi$  replaced by

$$\chi_F(q) = c_v + q r_K \tau + (1 - q) r_F \tau = (c_v + r_F \tau) + q (r_K - r_F) \tau.$$

Under Assumptions 1, 3 and 4<sub>F</sub>: (i)  $S$  is an open interval, non-empty if and only if  $V(q^*) > c_v + \bar{r} \tau$  with  $\bar{r} = q^* r_K + (1 - q^*) r_F$ , and then  $S = (q_1, q_2)$  with  $q_L \leq q_1 < q^* < q_2 \leq q_U$ ,

$$q_1 = \frac{c_v + r_F \tau + (1 - t) b}{s a + (1 - t) b - (r_K - r_F) \tau}, \quad q_2 = \frac{t b - c_v - r_F \tau}{t b + (1 - s) a + (r_K - r_F) \tau},$$

both denominators positive; the optimal policy is that of Theorem 1 with these end points. (ii)  $\tau < \tau'$  implies  $S(\tau') \subseteq S(\tau)$ . If  $V(q^*) \leq c_v$ , then

$S(\tau) = \emptyset$  for every  $\tau \geq 0$ . If  $\bar{r} > 0$  and  $V(q^*) > c_v$ , then  $S(\tau) \neq \emptyset$  if and only if  $\tau < \tau_F^*$ , where

$$\tau_F^* = \frac{V(q^*) - c_v}{q^* r_K + (1 - q^*) r_F}.$$

(iii) If moreover  $r_K > 0$ , then  $\tau_F^* = \tau^* q^* r_K / \bar{r} \leq \tau^*$ , with equality if and only if  $r_F = 0$ : the critical delay  $\tau^*$  of Theorem 2 is an upper bound on the critical delay when faults accrue cost during the delay.

#### 4.7. Populations of cases: what the prior changes

The verification region is a set of beliefs and does not depend on the prior. An operator facing a population of cases—for instance all the anomalies raised in one simulated market at one base rate of compromise—sees a distribution of passive posteriors, which the prior moves.

**Definition 5 (Verification mass, saving and recovered share of a cell).** An *order-flow model*  $c$  is a pair of order-flow laws  $P_K^c, P_F^c$  on a measurable space, with densities  $f_K^c, f_F^c$  with respect to a  $\sigma$ -finite measure  $\nu$  dominating both (e.g.  $\nu = P_K^c + P_F^c$ ). A *cell*  $(c, p)$  is an order-flow model  $c$  together with a prior  $p \in (0, 1)$ ; in Paper 1's grid,  $c$  is a market design in one regime and  $p$  is a prior of the grid. In the setting of Definition 4 with  $P_K^Y = P_K^c$ ,  $P_F^Y = P_F^c$  and prior  $p \in (0, 1)$ , write  $P_p = p P_K^c + (1 - p) P_F^c$  for the law of  $Y$  and

$$q_p(y) = \frac{p f_K^c(y)}{p f_K^c(y) + (1 - p) f_F^c(y)}$$

for the passive posterior (defined where the denominator is positive, a set of full  $P_p$ -measure). The *verification mass*, the *saving* and the *recovered share* of the cell  $(c, p)$  are

$$m_c(p) = P_p(q_p(Y) \in S), \quad D_c(p) = \mathbb{E}_p[(V - \chi)^+(q_p(Y))],$$

$$\rho_c(p) = \frac{D_c(p)}{\text{CoA}_c^*(p)},$$

with  $S$ ,  $V$ ,  $\chi$  as in Definition 3 and  $\text{CoA}_c^*(p)$  the passive cost of ambiguity of Definition 4 for this cell;  $\rho_c(p)$  is defined when  $\text{CoA}_c^*(p) > 0$ . Where  $f_K^c, f_F^c > 0$ ,  $\text{logit } q_p(y) = \text{logit } p + \Lambda_c(y)$  with  $\Lambda_c = \log(f_K^c/f_F^c)$ .

**Lemma 4 (Saving identity and low-prior limit).** Under Assumptions 1–4, in the setting of Definition 5: (i) *Saving.*  $D_c(p) = \text{CoA}_c^*(p) -$

$\text{CoA}_{v,c}^*(p) = R_Y^* - R_{Y,v}^*$ , where  $\text{CoA}_{v,c}^*(p)$  is the verify-first cost of ambiguity of Definition 4 for this cell; it is attained by the policy that verifies iff  $q_p(Y) \in S$ , then contains on  $Z = +$  and leaves on  $Z = -$ , and otherwise takes the passive action of Lemma 1 at  $q_p(Y)$ . Moreover  $0 \leq D_c(p) \leq \min\{\text{CoA}_c^*(p), V(q^*) m_c(p)\}$ , so  $0 \leq \rho_c(p) \leq 1$  where defined. (ii) *The saving vanishes at low priors.* For every order-flow model  $c$ ,  $D_c(p) \rightarrow 0$  as  $p \rightarrow 0$ . (iii) *The mass vanishes at low priors when  $q_1 > 0$ .* If  $S = \emptyset$ , then  $m_c \equiv 0$ . If  $S = (q_1, q_2) \neq \emptyset$  and  $q_1 > 0$  (equivalently,  $c_v > 0$  or  $t < 1$ ), then  $m_c(p) \rightarrow 0$  as  $p \rightarrow 0$ . (iv) *The exception.* If  $S \neq \emptyset$ ,  $c_v = 0$  and  $t = 1$  (so  $q_1 = 0$ ), then  $m_c(p) \rightarrow P_F^c(f_K^c(Y) > 0)$  as  $p \rightarrow 0$ . This limit is positive if and only if  $P_K^c$  and  $P_F^c$  are not mutually singular, and zero when they are (compromise and fault perfectly separable by order flow); the saving still vanishes, by (ii). No condition on the order-flow model is needed beyond Definition 5: the densities are finite, and  $f_F^c > 0$  holds  $P_F^c$ -almost surely by construction.

Part (ii) is immediate from part (i) and Corollary 2(ii). Part (iii) is not: as the prior falls, the passive posteriors move below  $q_1$  whenever the check has a fixed cost or can flag a fault, so the mass in the interval vanishes too. A low prior does not by itself make verification more valuable; what matters is whether the upper tail of the posteriors still reaches above  $q_1$ , which the theory does not decide. This is why the empirical question of Section 5 is posed per population. Their closed forms for a Gaussian order-flow model (Proposition 1, Supplementary Material, Section S1) are a calibration cross-check that decides no claim (Remark 2).

## 5. Exploratory empirical study on Paper 1's cells

This section reports an exploratory study on the order-flow models and priors of Paper 1 [1]; when each criterion was specified, and its status, are in Table 1. The grids, plausible set, policies, metrics and the primary test's outcome criterion were fixed before any quantity of the primary test was computed. The theorems of Section 4 are proved and are not tested here. The full design is in the Supplementary Material, Section S3, and its values in Table S1.

The study evaluates the rule, not a check. The simulated designs have no authorization variable, the check's  $(s, t)$  and Assumption 2 are inputs set on a grid, and verification outcomes enter only through their exact expectation; the data enter only through Paper 1's corrected posteriors. The study therefore bears on where a check with given characteristics would pay on those

posteriors, and through the agreement criterion on their calibration; it cannot bear on Assumption 2, on whether any existing check has the assumed characteristics, or on whether the simulated compromises would fail such a check.

### 5.1. Data, cells and the failure cells

Paper 1’s order-flow models are simulated with ABIDES, an agent-based discrete-event simulator of a limit order book market [19]. A *market design* is one way of injecting a compromise and a fault into the simulated market, run in two *regimes*: in the *summary-matched* regime the two injections are parameterised so that summary statistics of their order flow match, and in the *contrast* regime they are not. The data are Paper 1’s simulated order-flow features for its 5 profiles (D1, D1+, D2, D3, D4), from 4 injection designs; D1+ re-featurises the episodes of D1, so its cells are not independent of D1’s. On each profile and regime, Paper 1’s classifier is refitted exactly as Paper 1 fits it and applied to the held-out test instances, balanced between the causes ( $n_K = 300$ ,  $n_F = 300$ ). Its scores are moved from the training share  $\pi_{\text{tr}} = 0.5$  of compromises to a prior  $p$  by the standard prior-shift correction [20], and the corrected posteriors are the passive beliefs  $q(y)$  to which Lemma 2 applies the verification analysis.

A *cell* is a profile, a regime and a prior  $p \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$  of Paper 1’s grid (Definition 5); its quantities are expectations under  $P_p = p P_K^c + (1 - p) P_F^c$ , obtained by weighting the balanced test set. The *failure cells* are the cells in which Paper 1 reports that its classifier does not beat the best constant policy, always contain or always leave; they are read from Paper 1’s reported cost table, not re-derived, and listed in Table 2. They are the point-estimate exceptions of that table, a subset of the cells Paper 1 hands over to this paper: in the other cells of that set the classifier beats the best constant policy at the point estimate, by an amount Paper 1’s interval does not resolve. On those cells this study decides nothing, so its answer to the question Paper 1 leaves open is partial.

The primary loss table is the special case  $H_F = 0$  of Paper 1’s Definition D6, in the form of Section 3.1. Paper 1 itself classifies its cells with a measured, positive  $H_F$  that differs by profile and regime; with  $H_F = 0$ , the passive threshold  $q^*$  is the same in every cell, so the predicted region depends on a cell only through its corrected posteriors. The choice was fixed before any result. The measured- $H_F$  table is a second arm, reported for every criterion; re-deriving the failure cells by Paper 1’s point-estimate rule from the

same posteriors gives the same 6 cells under  $H_F = 0$  and under the measured table, and every outcome word is the same under both.

### 5.2. Grid, policies and intervals

The model involves the delay and the damage rate only through  $r_K\tau$ , so the grid has four axes in model units:  $s$ ,  $t$ , the cost share  $x = c_v/V_{\max}$  and the delay share  $y = r_K\tau/a$ , with  $V_{\max} = ab/(a+b)$  the peak value of a perfect, free and instantaneous check (Lemma 3). A non-empty interval then reads  $x + y < s + t - 1$ , independent of a cell's monetary scale. A grid point is *implausible* if its specificity is the highest level of the grid and its cost one of the two lowest levels of  $x$ , a near-free check that almost never flags a fault and so pushes  $q_1$  towards zero (Lemma 4(iv)); every other point is *plausible*. The levels are  $s \in \{0.7, 0.8, 0.9, 0.99\}$ ,  $t \in \{0.7, 0.8, 0.9, 0.99\}$ ,  $x \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5\}$  and  $y \in \{0, 0.01, 0.05, 0.1, 0.25, 0.5\}$ , giving 576 points of which 528 are plausible. The upper-tail levels are  $\alpha_{\text{in}} = 0.9$  and  $\alpha_{\text{out}} = 0.99$ , and  $\gamma = 0.95$ , so  $\bar{\sigma} = 0.9$ ; all were fixed before any quantile was computed, as choices, not derived from the sample size.

Five policies are compared on every cell and grid point: always contain; always leave; the passive classifier (Lemma 1); verify everything and follow the outcome; and targeted verification by the rule of Theorem 1 applied to the corrected posterior. Verification outcomes are not simulated: each policy's cost is its exact expectation over the outcome, so the only randomness is the resampling of the test set. Confidence intervals are bootstrap percentile intervals over test seeds, each resample keeping both episodes of each drawn seed; this unit replaced, by a rule fixed before the re-analysis, a first analysis that resampled test instances within each cause, reported beside it as superseded. False-positive and false-negative rates are reported beside every cost.

### 5.3. Primary exploratory test: targeted verification in the failure cells

The primary test asks, in each failure cell, over which region of  $(s, t, c_v, \tau)$  targeted verification beats the cell's best constant policy, and whether that region is the one predicted from Theorem 1 and the cell's corrected posteriors. Write  $\eta(c, g)$  for the expected cost of targeted verification minus that of the best constant policy at grid point  $g$ , with bootstrap interval  $[L, U]$ . When that policy is always leave, Lemma 1 and Lemma 4(i) predict the contrast  $-\Gamma(c, g)$ , with

$$\Gamma(c, g) = \mathbb{E}_p \left[ (a+b)(q - q^*)^+ + (V - \chi)^+(q) \right]$$

over the corrected posteriors (with  $(q^* - q)^+$  when it is always contain).

*Predicted region and agreement.* With  $Q_\alpha$  the  $\alpha$ -quantile of the corrected posterior under  $P_p$  and two upper-tail levels  $\alpha_{\text{in}} < \alpha_{\text{out}}$ , a grid point is *predicted in* if  $q_1 < Q_{\alpha_{\text{in}}}$  and  $P_p(q_1 < q < q_2) > 0$ , *predicted out* if  $q_1 \geq Q_{\alpha_{\text{out}}}$  or all mass above  $q_1$  lies at or above  $q_2$ , and in the *band* otherwise; band points are counted and enter no share. At a predicted-in point the result *agrees* if  $U < 0$ , *disagrees* if  $U \geq 0$  and  $L > -\Gamma$ , and is *unresolved* if the interval contains both 0 and  $-\Gamma$ . At a predicted-out point it agrees if  $U \geq 0$  and disagrees if  $U < 0$ . A cell's agreement share is agreeing over agreeing and disagreeing plausible points; a cell *agrees* if the share is at least  $\bar{\sigma} = 1 - 2(1 - \gamma)$ , with  $\gamma$  the confidence level, and is *unresolved* if more than half of its plausible predicted-in points are unresolved.

*Outcome.* Applied in this order: *refutes* if no plausible point of any failure cell has  $U < 0$  and the test has power (some failure cell has plausible predicted-in points, and in every such cell at least half of them disagree); *weakens* if some failure cell has a share below  $\bar{\sigma}$ , or  $U < 0$  only at implausible points; *inconclusive at this power* if no cell weakens or refutes but some cell is unresolved; *supports* if every failure cell agrees, none is unresolved, and  $U < 0$  at one plausible point of one failure cell at least. The last condition carries little evidence: by Lemma 4(i), at a nearly free and accurate check the plug-in saving is positive whenever some posterior mass lies above  $q_1$ , so it excludes only a region that is empty at every plausible point. The outcome is decided with the  $H_F = 0$  table; the measured- $H_F$  version is reported beside it and decides nothing.

The tested claim replaced an earlier one whose premise failed on Paper 1's cost table before any quantity of this study was computed; the test was run first on a provisional version of Paper 1's cells, where it weakened the prediction, and then with the same rules on Paper 1's final cells after Paper 1 refitted its classifiers (Supplementary Material, Section S3).

*Policy comparison.* If the corrected posteriors are calibrated, targeted verification is, in expectation, at least as cheap as every other policy on every cell (Lemma 4(i)). The comparison is read as weakening this if the interval of targeted verification minus some other policy lies above zero at some grid point of some cell, which would indicate miscalibrated posteriors, and as refuting it if targeted verification never beats the best constant policy in the failure cells.

#### 5.4. Results

Table 1 lists every criterion of the study side by side: the check of the computation, the two exploratory outcomes of this section with the primary test’s sensitivity analysis, and two pre-registered tests that were designed afterwards and stopped at their own pre-checks. None of them is confirmatory. Neither pre-registered test could reach its required power on a feasible sample, so neither was run (Section 6.1).

*Primary exploratory test.* On the 6 failure cells of Paper 1, the outcome of the primary test is *supports* under its pre-stated criterion and would be *weakens* under the post-hoc predicted-in reading; it is exploratory: the test was specified after Paper 1’s cost table had been examined, so it is not a pre-registered confirmation. Every failure cell reaches the minimum agreement share  $\bar{\sigma} = 0.9$  over the 528 plausible grid points (smallest share 0.961), no cell is unresolved, and in every failure cell targeted verification beats always leaving, the best constant policy there, at plausible grid points (Table 2). That share counts predicted-out points, where the tail above  $q_1$  is too thin for a saving to be resolved, so they can rarely disagree. Restricted to predicted-in points, the agreement share lies between 0.762 and 0.995 across the failure cells and falls below  $\bar{\sigma}$  in 2 of them, the failure cells of D2 in the summary-matched regime at the two lowest priors  $p$  (Table 2, columns In agree and In share). Its empirical content is that the region where targeted verification beats always leaving is non-empty at plausible grid points in every failure cell. The agreement of that region with the predicted one is, since Theorem 1 and Lemma 4(i) are proved, a check of the calibration of the corrected posteriors in these cells, not evidence for the theorems. Each failure cell keeps disagreeing points, and in the contrast regime of D1 many points (341) fall in the band between the two quantile levels and enter no share.

The superseded within-cause analysis gave the same outcome, with the predicted-in reading below  $\bar{\sigma}$  in 1 cell. The measured cost of leaving a fault gives the same outcome (smallest share 0.962; a sensitivity analysis that decides nothing). Details, and the same rule applied after the fact to the cells Paper 1 hands over that are not failure cells, are in the Supplementary Material, Section S4.

*Policy comparison.* With family-wise control (max-T over the plausible points of a cell, Holm across the cell-comparator pairs), no cell keeps a significant loss of targeted verification under either loss table, and in every failure cell

Table 1: Every criterion of the empirical study, as decided by its pre-stated rule on Paper 1’s final cells (seed-paired bootstrap, the method of record), beside the superseded within-cause bootstrap, the post-hoc predicted-in reading (decides nothing; Table 2) and the same rules on Paper 1’s provisional cells, superseded when Paper 1 refitted its classifiers (–: not applicable). Policy comparison: the row gives the per-point criterion without multiplicity control; with family-wise control, the method of record for this row, the outcome is *inconclusive at this power* (Section 5.4).

| Criterion                                                                                          | Status                                                                                                       | Decided / not run | Outcome (seed-paired, record)          | Within cause (super-seded) | Predicted in reading (post hoc) | Provisional basis | Decides                                                  | Limit                                                                                                                           |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------|----------------------------|---------------------------------|-------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Interval and critical delay: closed form against numerical search                                  | check of the computation (a theorem)                                                                         | decided           | cross-check passed                     | –                          | –                               | –                 | nothing (a theorem; mismatch is an error of computation) | a check of the computation, a not evidence                                                                                      |
| Primary exploratory (Definition D6)                                                                | ex- exploratory; test specified after Paper 1’s table was examined                                           | decided           | supports                               | supports                   | weakens                         | weakens           | the outcome of the primary test (exploratory)            | counts predicted-out points; predicted-in reading <i>weakens</i> in 2 cells; checks the calibration of the corrected posteriors |
| Primary measured (sensitivity) leaving a fault the primary test                                    | test, exploratory (sensitivity) test                                                                         | decided           | supports                               | supports                   | weakens                         | weakens           | nothing (sensitivity)                                    | predicted-in reading <i>weakens</i> in 1 cells                                                                                  |
| Policy comparison                                                                                  | exploratory; criterion stated before any cost existed, per-point reading confirmed after provisional results | decided           | weakens                                | weakens                    | –                               | weakens           | the policy-comparison criterion                          | no multiplicity control (brief B006)                                                                                            |
| Pre-registered confirmation of the primary test (brief B002)                                       | pre-registered; before any fresh instance                                                                    | not run           | inconclusive at this power (pre-check) | –                          | –                               | –                 | the primary test, confirmed                              | power check failed in one cell; not run                                                                                         |
| Pre-registered test of the calibration mechanism (Proposition 2) and of recalibration (brief B003) | pre-registered; before any fresh instance                                                                    | not run           | inconclusive at this power (pre-check) | –                          | –                               | –                 | the mechanism of the policy-comparison                   | no cell passed the registered conclusion floor; not run                                                                         |

Table 2: Primary exploratory test per failure cell (seed-paired bootstrap, the method of record), with the  $H_F = 0$  special case of Paper 1’s Definition D6 and, below it, with the measured cost of leaving a fault (sensitivity, decides nothing). Counts of plausible grid points predicted in, out and in the band; agreeing, disagreeing and unresolved points; the agreement share against  $\bar{\sigma} = 0.9$  (the pre-stated criterion); beside it, post hoc, the predicted-in points that agree and their share of the decided predicted-in points (–: none decided). Column Impl.  $U < 0$ : whether targeted verification beats the baseline at some implausible point. Last three columns: the first term of  $\Gamma$  (the plug-in saving of the passive rule over the best constant policy; one value per cell), and the range over the plausible grid points of  $\Gamma$  and of its second term (the plug-in saving of targeted verification over the passive rule), in Paper 1’s monetary units. Designs as in Paper 1 (D1, D2); C: contrast, SM: summary-matched regime.

| Cost matrix    | Design                   | Regime | $p$  | Baseline | In  | Out | Band | Agree | Disagree | Unres. | Share | In agree | In share | Agrees | Unres. cell | Impl. $U < 0$ | $\Gamma$ 1st term | $\Gamma$ [min, max] | $\Gamma$ 2nd term [min, max] |
|----------------|--------------------------|--------|------|----------|-----|-----|------|-------|----------|--------|-------|----------|----------|--------|-------------|---------------|-------------------|---------------------|------------------------------|
| $H_F = 0$ (D6) | D1                       | C      | 0.05 | leave    | 120 | 67  | 341  | 182   | 2        | 3      | 0.989 | 115      | 0.983    | yes    | no          | yes           | 1,723             | [1,723, 26,924]     | [0, 25,201]                  |
|                | D2                       | C      | 0.05 | leave    | 103 | 214 | 211  | 296   | 1        | 20     | 0.997 | 82       | 0.988    | yes    | no          | yes           | 1,878             | [1,878, 22,689]     | [0, 20,811]                  |
|                | D2                       | SM     | 0.05 | leave    | 52  | 417 | 59   | 449   | 10       | 10     | 0.978 | 32       | 0.762    | yes    | no          | yes           | 0                 | [0, 20,619]         | [0, 20,619]                  |
|                | D2                       | SM     | 0.10 | leave    | 162 | 242 | 124  | 369   | 15       | 20     | 0.961 | 127      | 0.894    | yes    | no          | yes           | 0                 | [0, 61,022]         | [0, 61,022]                  |
|                | D2                       | SM     | 0.20 | leave    | 377 | 69  | 82   | 402   | 15       | 29     | 0.964 | 333      | 0.957    | yes    | no          | yes           | 337               | [337, 141,915]      | [0, 141,579]                 |
|                | D2                       | SM     | 0.30 | leave    | 461 | 67  | 0    | 509   | 2        | 17     | 0.996 | 442      | 0.995    | yes    | no          | yes           | 6,137             | [6,137, 223,073]    | [0, 216,936]                 |
|                | pooled (decides nothing) |        |      |          |     |     |      | 2207  | 45       | 0.980  |       |          |          |        |             |               |                   |                     |                              |
| measured $H_F$ | D1                       | C      | 0.05 | leave    | 129 | 67  | 332  | 188   | 7        | 1      | 0.964 | 121      | 0.945    | yes    | no          | yes           | 1,922             | [1,922, 27,122]     | [0, 25,200]                  |
|                | D2                       | C      | 0.05 | leave    | 103 | 213 | 212  | 295   | 1        | 20     | 0.997 | 82       | 0.988    | yes    | no          | yes           | 1,879             | [1,879, 22,695]     | [0, 20,815]                  |
|                | D2                       | SM     | 0.05 | leave    | 52  | 416 | 60   | 448   | 10       | 10     | 0.978 | 32       | 0.762    | yes    | no          | yes           | 0                 | [0, 20,627]         | [0, 20,627]                  |
|                | D2                       | SM     | 0.10 | leave    | 162 | 242 | 124  | 369   | 13       | 22     | 0.966 | 127      | 0.907    | yes    | no          | yes           | 0                 | [0, 61,030]         | [0, 61,030]                  |
|                | D2                       | SM     | 0.20 | leave    | 377 | 69  | 82   | 404   | 16       | 26     | 0.962 | 335      | 0.954    | yes    | no          | yes           | 341               | [341, 141,922]      | [0, 141,581]                 |
|                | D2                       | SM     | 0.30 | leave    | 461 | 67  | 0    | 509   | 5        | 14     | 0.990 | 442      | 0.989    | yes    | no          | yes           | 6,187             | [6,187, 223,079]    | [0, 216,892]                 |
|                | pooled (decides nothing) |        |      |          |     |     |      | 2213  | 52       | 0.977  |       |          |          |        |             |               |                   |                     |                              |

targeted verification remains significantly cheaper than the best constant policy; the policy comparison is *inconclusive at this power*. This verdict is partly forced by the design of the re-analysis: with the number of resamples used, the smallest attainable bootstrap p-value lies above Holm’s first-step level, so no cell-comparator pair could be rejected in that direction, whatever the data. Without control, targeted verification is significantly costlier than the passive classifier at some plausible point in 14 success cells (the earlier, uncontrolled reading, superseded; between 1 and 315 of the 528 plausible points, depending on the cell, Supplementary Material, Table S9; 10 of 34 without the D1+ cells, which re-use the episodes of D1). In those cells the realised saving over the passive rule is negative while the saving predicted from the corrected posteriors is positive, which the criterion named in advance as the sign of miscalibrated corrected posteriors; miscalibration is a hypothesis these cells suggest, not a finding of this study. Proposition 2 (Section 6.4) names a candidate mechanism, not tested here (descriptive detail in the Supplementary Material, Section S4).

## 6. Discussion

### 6.1. What the empirical study shows and does not show

The study of Section 5 does not test the theorems of Section 4, which are proved. What it can test is whether, on the corrected posteriors of a set of simulated market designs, the region where targeted verification beats the best constant policy is the one Theorem 1 predicts from them, which bears first on their calibration (Section 6.4). A confirmation of the primary test needs a pre-registered test on fresh instances, decided on the predicted-in points under both loss tables. Such a test was designed; under its pre-registered power check, no sample size in the costed range reached the required power in one of the failure cells, the D2 cell of the summary-matched regime at the second-lowest failure prior. The check estimates the probability that the predicted-in share reaches  $\bar{\sigma}$ . That share is taken over decided points, and the set of decided points grows with the sample: points whose interval at the present size contains both 0 and  $-\Gamma$  are unresolved and enter no share, and in that cell many of them are decided as disagreeing at larger sizes. The mean share falls as the sample grows and lies below  $\bar{\sigma}$  at the largest costed size, so the probability of meeting the criterion falls with the sample. By its pre-registered rule the outcome is *inconclusive at this power*, and the test was not run; read with the mechanism above, a larger sample would most likely have weakened the prediction in that cell. It would not show an empty region: targeted verification beats always leaving there at plausible points. The verification parameters are a grid in model units, not measurements of a particular gateway check, so the study says where a check would pay if it had given characteristics, not whether any existing check has them. Evidence on a real check, its accuracy and the fit of Assumption 2 needs data on that check, which this paper does not have.

### 6.2. When the separate channel fails

Every result of Section 4 that involves order flow rests on Assumption 2; if the outcome of the check depends on the order flow given the cause, Lemma 2 does not hold, and the fit of the assumption has to be argued for each verification procedure. Proposition 3 (proof in the Supplementary Material, Section S1) weakens what has to be argued to lower bounds on the check's accuracy that hold whatever the order flow.

**Proposition 3 (Robust verification region under a relaxed separate channel).** Under Assumptions 1, 3 and 4, replace Assumption 2 by

the following: given the order-flow observation  $Y = y$ , the verification outcome has sensitivity  $s(y) = P(Z = + \mid M = K, Y = y)$  and specificity  $t(y) = P(Z = - \mid M = F, Y = y)$ , measurable in  $y$ , with  $s(y) \geq s_{\text{lo}}$  and  $t(y) \geq t_{\text{lo}}$  for every  $y$ , where  $s_{\text{lo}}, t_{\text{lo}} \in [0, 1]$  and  $s_{\text{lo}} + t_{\text{lo}} > 1$ . Write  $V_{s,t}$  and  $S_{s,t}$  for the value of information and the verification region of Definition 3 with characteristics  $(s, t)$ , and  $q(y) = P(K \mid Y = y)$  for the *calibrated* posterior: the true conditional probability of compromise given the order-flow observation, not a classifier's score (a miscalibrated score is the case of Proposition 2). Then: (i)  $V_{s,t}(q)$  is nondecreasing in  $s$  and in  $t$  for every  $q \in [0, 1]$ ; hence  $S_{s_{\text{lo}}, t_{\text{lo}}} \subseteq S_{s(y), t(y)}$  for every  $y$ ; (ii)  $P(K \mid Y = y, Z = z)$  is the update of Definition 2 applied to  $q(y)$  with characteristics  $(s(y), t(y))$ ; (iii) the *robust policy*, which verifies if and only if  $q(Y) \in S_{s_{\text{lo}}, t_{\text{lo}}}$ , then contains on  $Z = +$  and leaves on  $Z = -$ , and otherwise takes the passive action of Lemma 1 at  $q(Y)$ , has, conditional on  $Y = y$ , an expected cost below that of the passive rule by  $(V_{s(y), t(y)} - \chi)(q(y)) \geq (V_{s_{\text{lo}}, t_{\text{lo}}} - \chi)(q(y)) > 0$  if  $q(y) \in S_{s_{\text{lo}}, t_{\text{lo}}}$ , and equal to it otherwise. It is never worse than the passive rule at any  $y$ , and it uses only the lower bounds  $(s_{\text{lo}}, t_{\text{lo}})$ , not  $s(\cdot)$  and  $t(\cdot)$ .

What it does not cover is a check whose accuracy can fall to chance, for instance an adversary who also controls the holder's second factor.

### 6.3. One check, then act

The rule is derived for a single check followed by a terminal action (Assumption 3), with the delay charged in full and no order flow used while the check runs (Assumption 4). The fault rate  $r_F$  of Corollary 4 is not measured, and the sensitivity analysis keeps  $r_F = 0$ . Repeated checks, graded responses and evidence that keeps arriving during the delay make the problem sequential, which this paper does not treat.

### 6.4. Posteriors and calibration

The saving identity of Lemma 4 holds for the true posterior; applied to classifier scores it holds only as far as the corrected scores are calibrated, so a gap between the plug-in and the realised saving beyond sampling error points to miscalibration, not to a failure of the lemma. For a miscalibrated score Proposition 2 (proof in the Supplementary Material, Section S1) replaces the lemma.

**Proposition 2 (Plug-in rules on a miscalibrated score).** Under Assumptions 1, 3 and 4, let  $S = (q_1, q_2) \neq \emptyset$  (Theorem 1), and let  $\hat{Q}$  be a  $[0, 1]$ -valued score computed from the order-flow observation (for example a prior-

shift-corrected classifier posterior), jointly distributed with the cause  $M$ , with the verification outcome  $Z$  conditionally independent of  $\hat{Q}$  given  $M$  (as under Assumption 2). Write  $\eta(\hat{q}) = P(M = K \mid \hat{Q} = \hat{q})$  for its calibration map (a version of the conditional probability), and  $B_1 = (q_1, q^*]$ ,  $B_2 = (q^*, q_2)$ . The *passive plug-in rule* leaves if  $\hat{Q} \leq q^*$  and contains otherwise; the *targeted plug-in rule* verifies if  $\hat{Q} \in S$ , then contains on  $Z = +$  and leaves on  $Z = -$ , and otherwise acts as the passive plug-in rule. Let

$$D_1 = s a + (1 - t) b - r_K \tau, \quad D_2 = t b + (1 - s) a + r_K \tau,$$

the denominators of  $q_1$  and  $q_2$  in Theorem 1, both positive. Then: (i) conditional on  $\hat{Q} = \hat{q}$ , the expected cost of the passive plug-in rule minus that of the targeted one is  $D_1 (\eta(\hat{q}) - q_1)$  if  $\hat{q} \in B_1$ ,  $D_2 (q_2 - \eta(\hat{q}))$  if  $\hat{q} \in B_2$ , and 0 otherwise; (ii) hence at a score  $\hat{q} \in S$  the targeted rule is strictly cheaper than the passive rule if and only if  $\eta(\hat{q}) > q_1$  when  $\hat{q} \leq q^*$ , and if and only if  $\eta(\hat{q}) < q_2$  when  $\hat{q} > q^*$ : a score in  $B_1$  may overstate the probability of compromise by less than  $\hat{q} - q_1$ , and a score in  $B_2$  may understate it by less than  $q_2 - \hat{q}$ , and verifying there still costs less than the passive rule; (iii) (*band-frequency identity*) the expected cost of the passive plug-in rule minus that of the targeted one is

$$D_1 [P(M = K, \hat{Q} \in B_1) - q_1 P(\hat{Q} \in B_1)] + D_2 [q_2 P(\hat{Q} \in B_2) - P(M = K, \hat{Q} \in B_2)];$$

(iv) if  $\hat{Q}$  is calibrated ( $\eta(\hat{Q}) = \hat{Q}$  almost surely), this difference is  $\mathbb{E}[(V - \chi)^+(\hat{Q})] \geq 0$ , the saving of Definition 5 and Lemma 4(i).

It suggests, though it does not state, recalibration of the posterior inside the band as a remedy; whether recalibration removes the uncontrolled losses of the policy comparison is an empirical question this paper leaves open.

## 7. Conclusion

Under the assumption that a check reading authorization is conditionally independent of order flow given the cause, verification is not bound by what order flow can reveal, and it composes with any passive posterior. Transposing the threshold approach, a check pays exactly on an interval of beliefs around the passive containment threshold, and because a true compromise keeps doing harm while the check runs, that interval vanishes at a critical delay in closed form. The interval does not depend on the prior; the saving it brings to a population of cases does. As supporting extensions, the

separate-channel assumption can be weakened to lower bounds on the check’s accuracy, and the interval and the critical delay carry over to a check with any output.

Section 5 specifies an exploratory test of where targeted verification beats the best constant policy in the cells of Paper 1 where the fitted classifier does no better. Both of its outcomes are exploratory. Evaluated on the corrected posteriors of those cells, targeted verification beats that policy at plausible check parameters, and the primary test reads *supports* under its pre-stated criterion, which counts mostly points where no saving is predicted, and *weakens* in 2 cells under a post-hoc reading restricted to the points where a saving is predicted; either reading checks the calibration of the corrected posteriors. The policy comparison is *inconclusive at this power* after family-wise control; without control, it was weakened in some of the cells where that classifier succeeds, which suggests (but does not establish) miscalibrated corrected posteriors there. No pre-registered confirmation has been run; Section 6.1 states what the study shows and does not show.

## Supplementary material

The Supplementary Material contains the proofs of the results of Section 4 (Section S1), the results of Paper 1 used here (Section S2), the full design of the empirical study (Section S3), its further results and secondary tables (Section S4), and further remarks on Section 4 (Section S5).

## Data and code availability

The code that computes the thresholds, the critical delay and the empirical test of Section 5.4, the experiment configurations, the provenance register and the result files from which every number in this article is generated are in the project’s public repository, <https://github.com/mejrissi/Thesis>, with the provenance register (`experiments/provenance_register.csv`) and the validation report (`docs/validation_report.md`). A tagged release is made at submission and an archival copy with a persistent identifier on acceptance. The data of the empirical test are the posteriors of the companion study [1], whose deposit they share.

## **CRedit authorship contribution statement**

**Chiheb Mejrissi:** Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

## **Declaration of competing interest**

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## **Funding**

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

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